TWO WHEELER SAFETY DETECTION SYSTEM

PROFESSOR: Deepak Gangadharan Sir

Teaching Assitant: Arihanth Jain Sir

Real-Time IoT Safety Monitoring with Live Web Dashboard

The Worst Generation

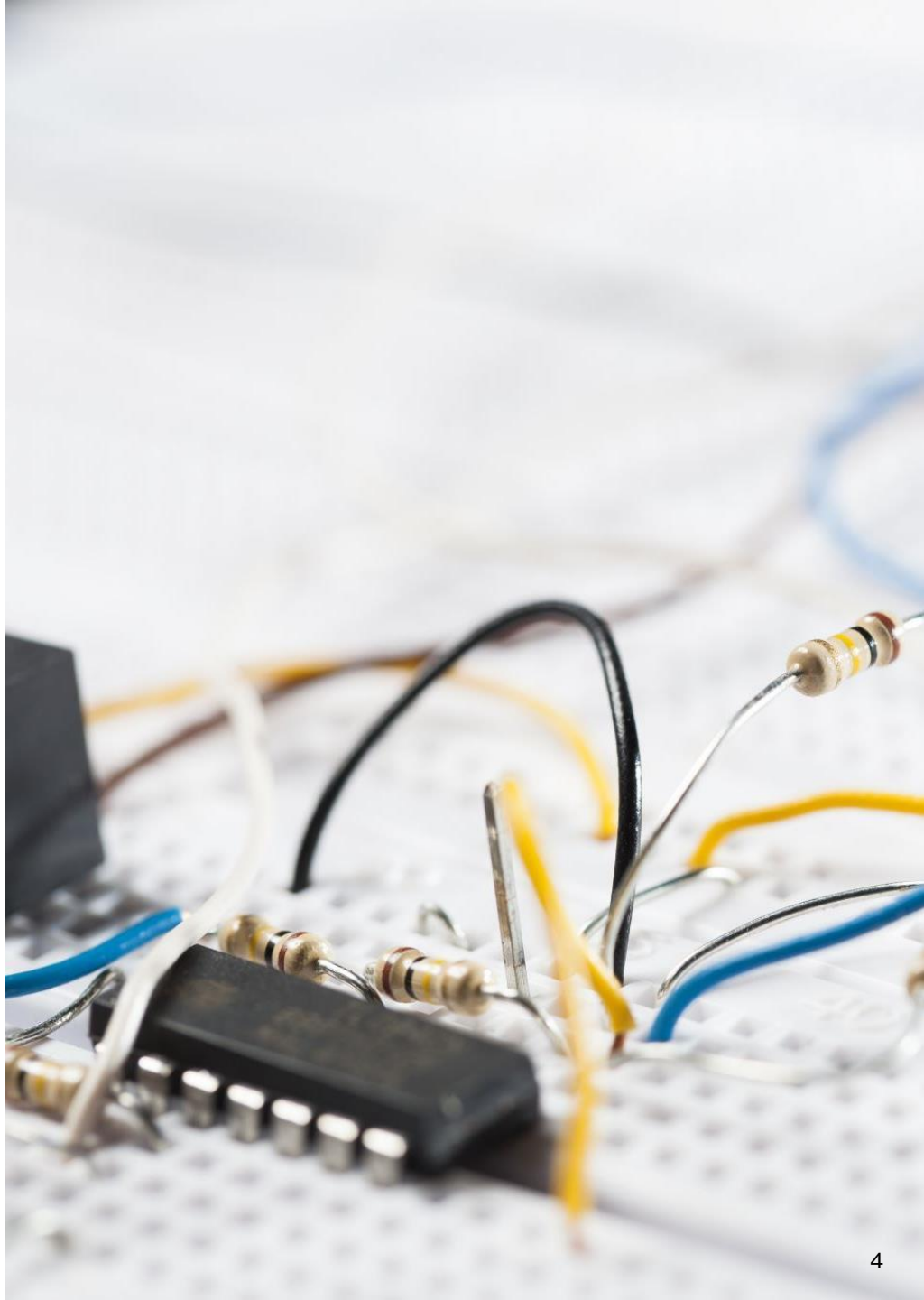
1. Santosh
2. Tilak
3. Naveen
4. Ravi

PROJECT OVERVIEW

- **What We Built:**
- Real-time safety monitoring for two-wheelers using IoT + AI
- Live web dashboard for instant visualization
- Safety Index scoring (0-10 scale)
- **Key Features:**
- Motion tracking & event detection
- Pothole detection with camera
- Real-time graphs and alerts
- Automatic safety scoring

HARDWARE COMPONENTS

- **Complete IoT System:**
- **Raspberry Pi 4** - Central processor, runs AI models
- **MPU6050 IMU** - 3-axis acceleration + gyroscope (104 Hz)
- **GPS Module** - Speed and location tracking
- **Pi Camera** - Road image capture for pothole detection
- **Custom 3D Printed Box** - Designed at Makers Lab, houses all hardware securely, mounted on scooter

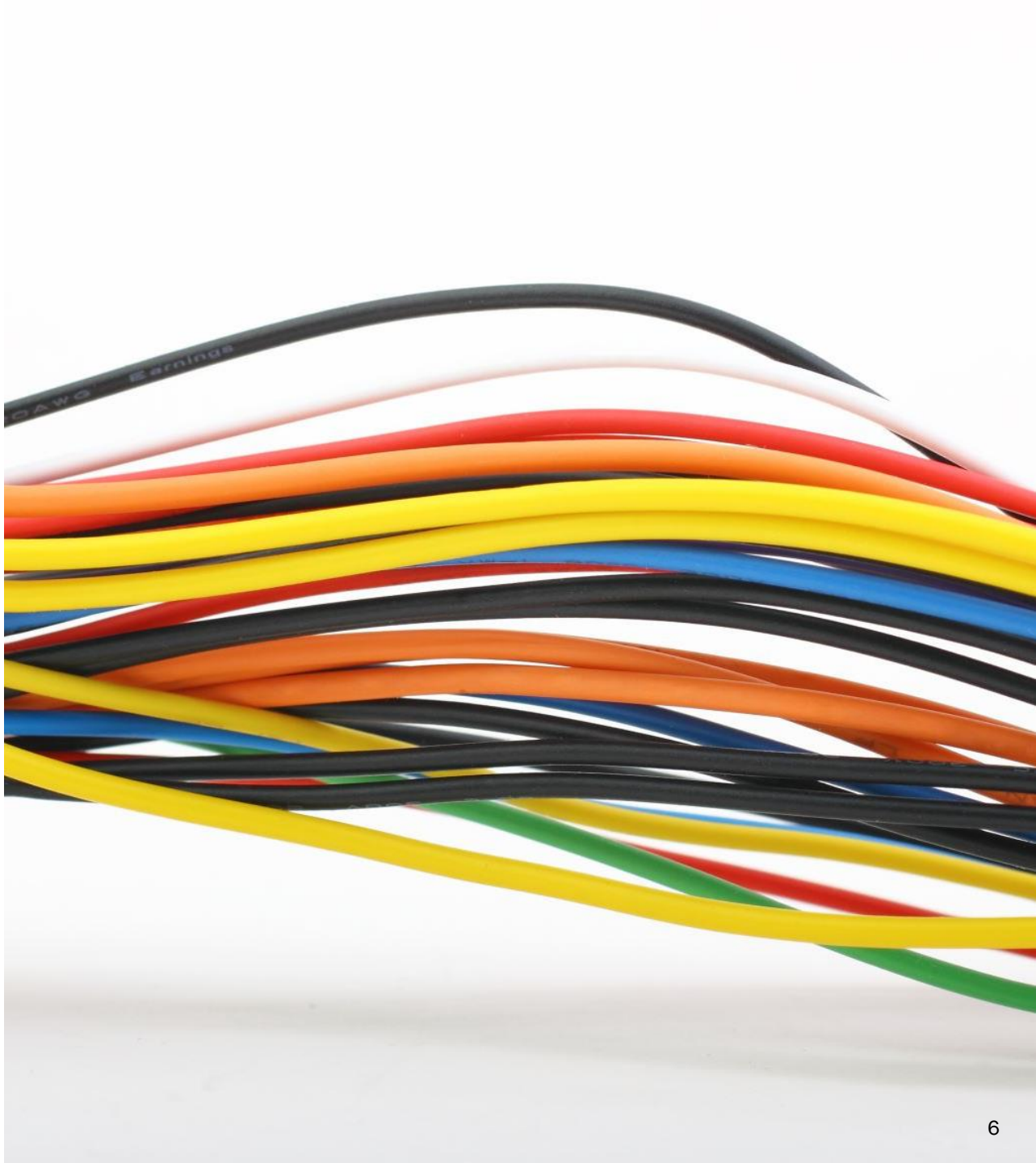


SENSOR CONNECTIONS TO RASPBERRY PI

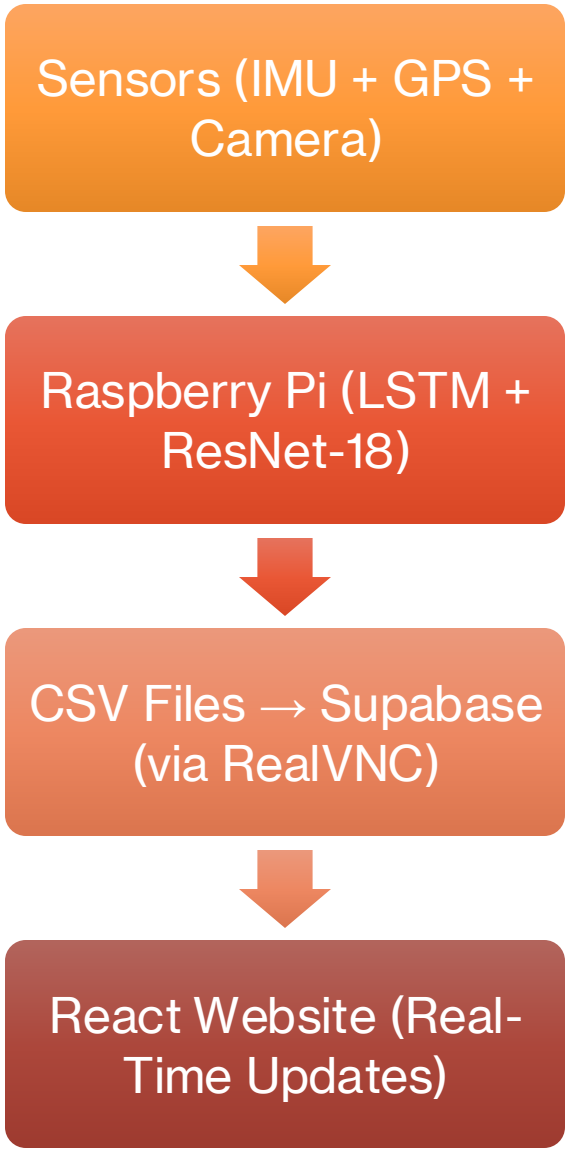
- **MPU6050 IMU → Raspberry Pi:**
- VCC → 3.3V (Pin 1)
- GND → Ground (Pin 6)
- SDA → GPIO 2 (Pin 3) - I2C Data
- SCL → GPIO 3 (Pin 5) - I2C Clock
- Protocol: I2C at 400kHz, 104Hz sampling rate
- **GPS Module → Raspberry Pi:**
- VCC → 5V (Pin 2) • GND → Ground (Pin 14)
- TX → GPIO 15 (Pin 10) - UART Receive
- RX → GPIO 14 (Pin 8) - UART Transmit
- Protocol: UART serial, 1Hz position updates

SENSOR CONNECTIONS TO RASPBERRY PI

- **Pi Camera → Raspberry Pi:**
- Connected via CSI (Camera Serial Interface) ribbon cable
- Direct connection to camera port on Raspberry Pi
- 1080p capture, triggered every 2 seconds
- **Power Supply:**
- 5V/3A adapter for Raspberry Pi
- Battery backup for 2+ hours of operation
- All sensors powered through Pi's GPIO pins



SYSTEM ARCHITECTURE

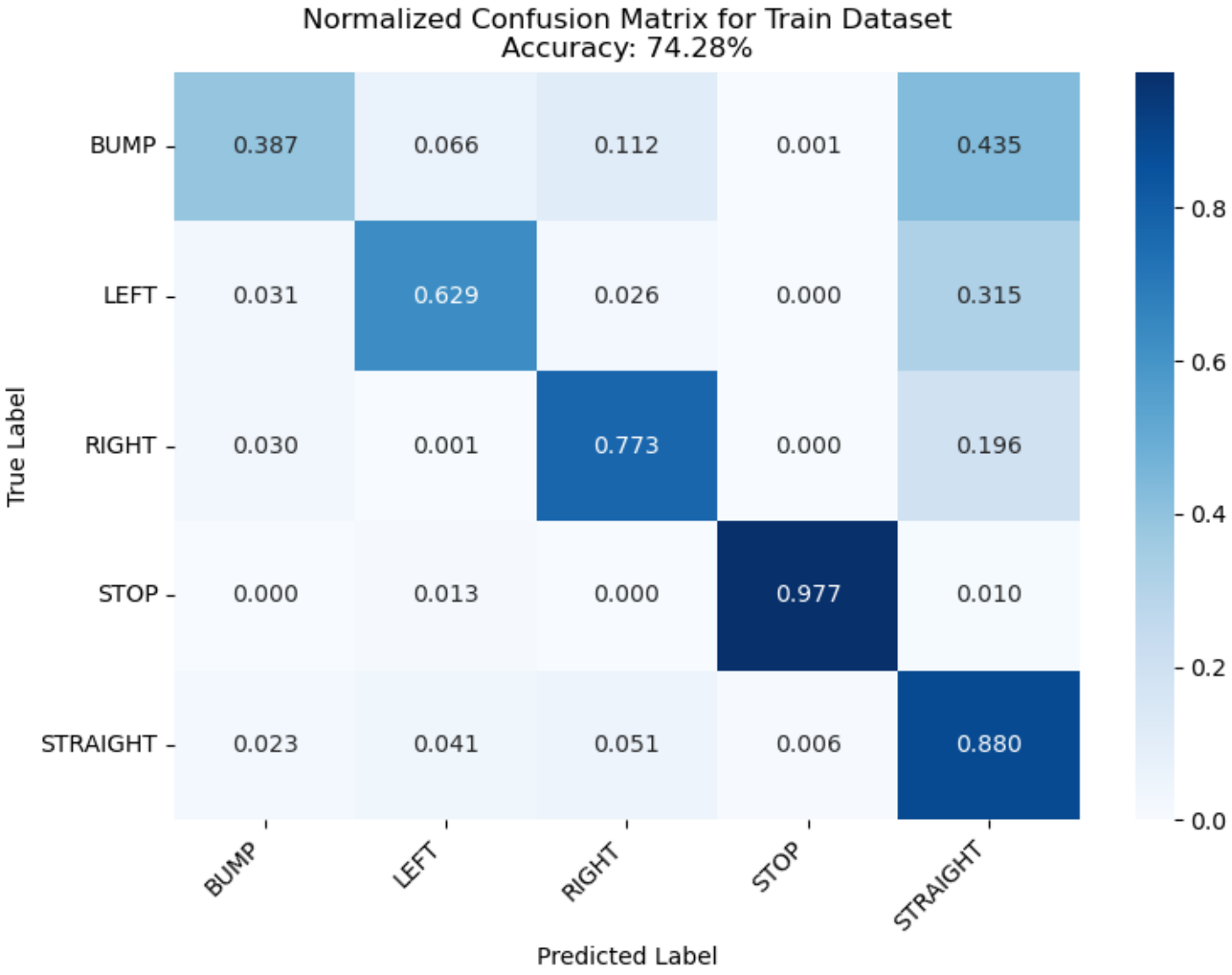


ML MODELS

- LSTM - Event Detection:
- Detects: Turns, Stops, Straights, Bumps
- Updates dashboard timeline instantly
- ResNet-18 - Pothole Detection:
- Pre-trained CNN for computer vision
- Instant alerts with pothole images on dashboard

LSTM MODEL DETAILS

- **Architecture**: 2 LSTM layers (128 units each) + Dense layers
- **Input:** 50-sample sliding window (6 sensor channels)
- **Training**: 20,000+ labeled sequences, 94% accuracy
- **Inference Time**: 15ms per prediction
- **Output Classes**: Turn Left/Right, Hard Brake, Bump, Straight, Idle

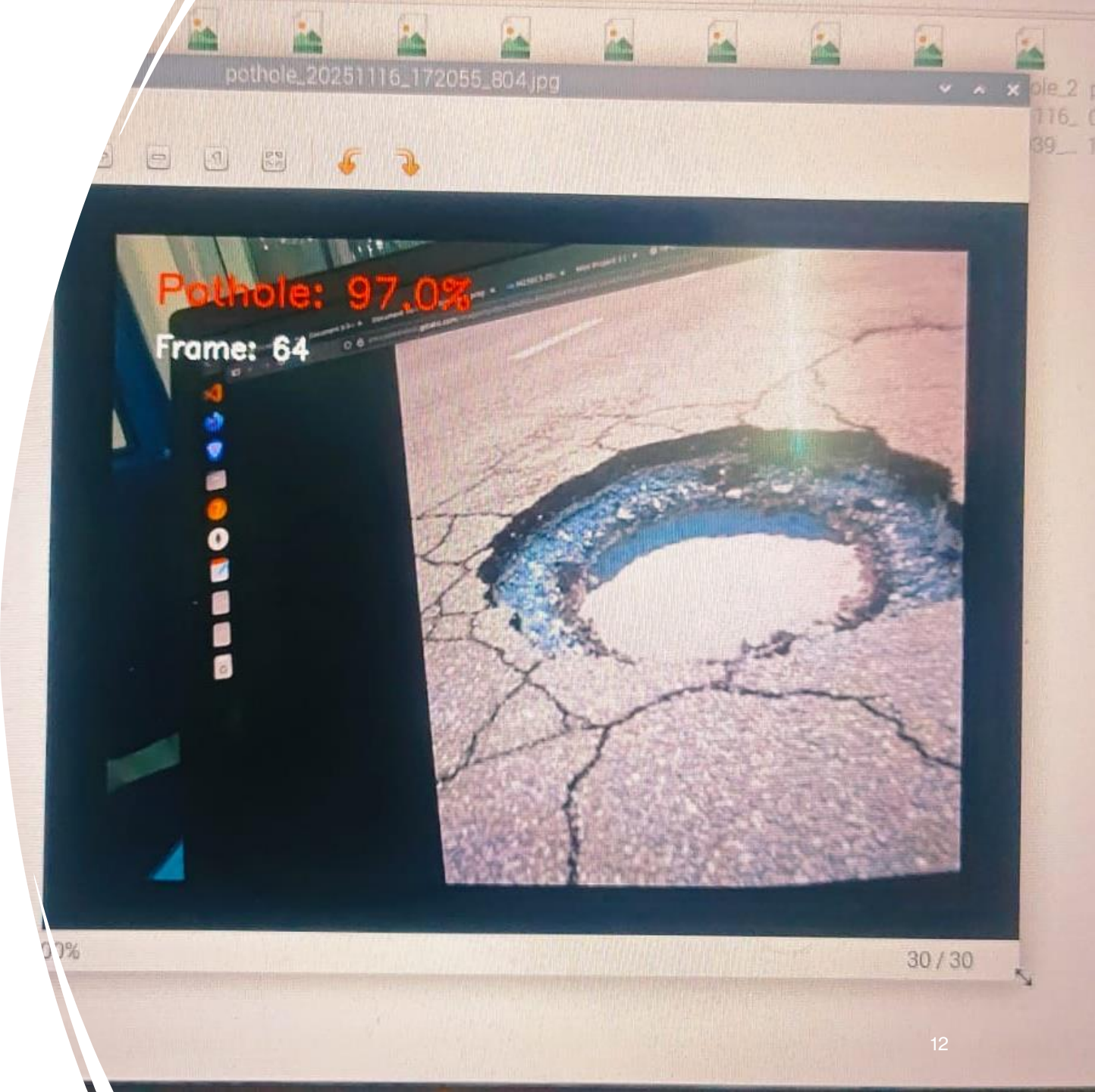


LSTM-MODEL ACCURACY

EVENT DETECTION & REAL-TIME DASHBOARD UPDATES

- **How LSTM Detects Events:**
- Analyzes patterns in acceleration + gyroscope data
- Sliding window continuously monitors last 0.48 seconds
- Recognizes unique motion signatures for each event
- **5 Detected Events:**
- 1. Straight - Steady forward motion, minimal rotation
- 2. Turn Left - Negative gyroscope Z-axis rotation
- 3. Turn Right - Positive gyroscope Z-axis rotation
- 4. Bump - Sharp vertical acceleration spike
- 5. stop - Near-zero motion on all axes
- Real-Time Dashboard:
- Events appear on timeline instantly with timestamp, color-coded markers, and severity indicators.

POTHOLE DETECTION



POTHOLE DETECTION PIPELINE

ResNet-18: Pre-trained on ImageNet, fine-tuned on road dataset

Image Processing: 224x224 resize, normalization

Detection Confidence: Threshold at 70%

Post-Processing: Filters false positives from shadows

Performance: 85% precision, 92% recall

SAFETY INDEX FORMULA

- **Weighted Score (0-10):**
- Safety Index = Acceleration (20%) + Gyroscope (20%) + Event Smoothness (25%) + Pothole Avoidance (20%) + Speed Consistency (15%)
- **Key Metrics:**
- Harsh acceleration: >0.6g
- Aggressive turns: >60°/s
- Each pothole: -0.5 points
- Speed variance penalties
- **Rating Scale:**
- 9-10: Excellent
- 8-9: Very Good
- 7-8: Good
- Below 7: Needs improvement

SAFETY SCORE BREAKDOWN

- **Example Calculation (14.7 min ride):**
- 1. **Acceleration Score**: 7.8/10 (Weight: 20%)
- Base score: 10.0
- 154 harsh acceleration events detected (>0.6g threshold)
- Penalty: -0.014 per event = -2.2 points
- Final component: 7.8/10
- 2. **Gyroscope Score**: 6.9/10 (Weight: 20%)
- Base score: 10.0
- 2,908 aggressive turns detected (>60°/s threshold)
- Penalty: -0.001 per aggressive turn = -3.1 points
- Final component: 6.9/10
- 3. **Event Smoothness**: 8.1/10 (Weight: 25%)
- Analyzes transitions between events
- Smooth gear shifts, gradual braking rewarded
- Jerky movements, sudden stops penalized
- Final component: 8.1/10

SAFETY SCORE BREAKDOWN

- 4. Pothole Avoidance: 9.0/10 (Weight: 20%)
- • 3 potholes encountered during ride
- • Penalty: -0.5 points per pothole = -1.5 point
- • Final component: 9.0/10
- 5. Speed Consistency: 8.4/10 (Weight: 15%)
- • Standard deviation of speed analyzed
- • Frequent acceleration/deceleration penalized
- • Smooth, steady speed rewarded
- • Final component: 8.4/10
- Final Safety Index: (7.8×0.20) + (6.9×0.20) + (8.1×0.25) + (9.0×0.20) + (8.4×0.15) = 7.83/10 (GOOD)

REAL-TIME DATA FLOW

How Data Reaches Dashboard:

1. Raspberry Pi collects sensor data

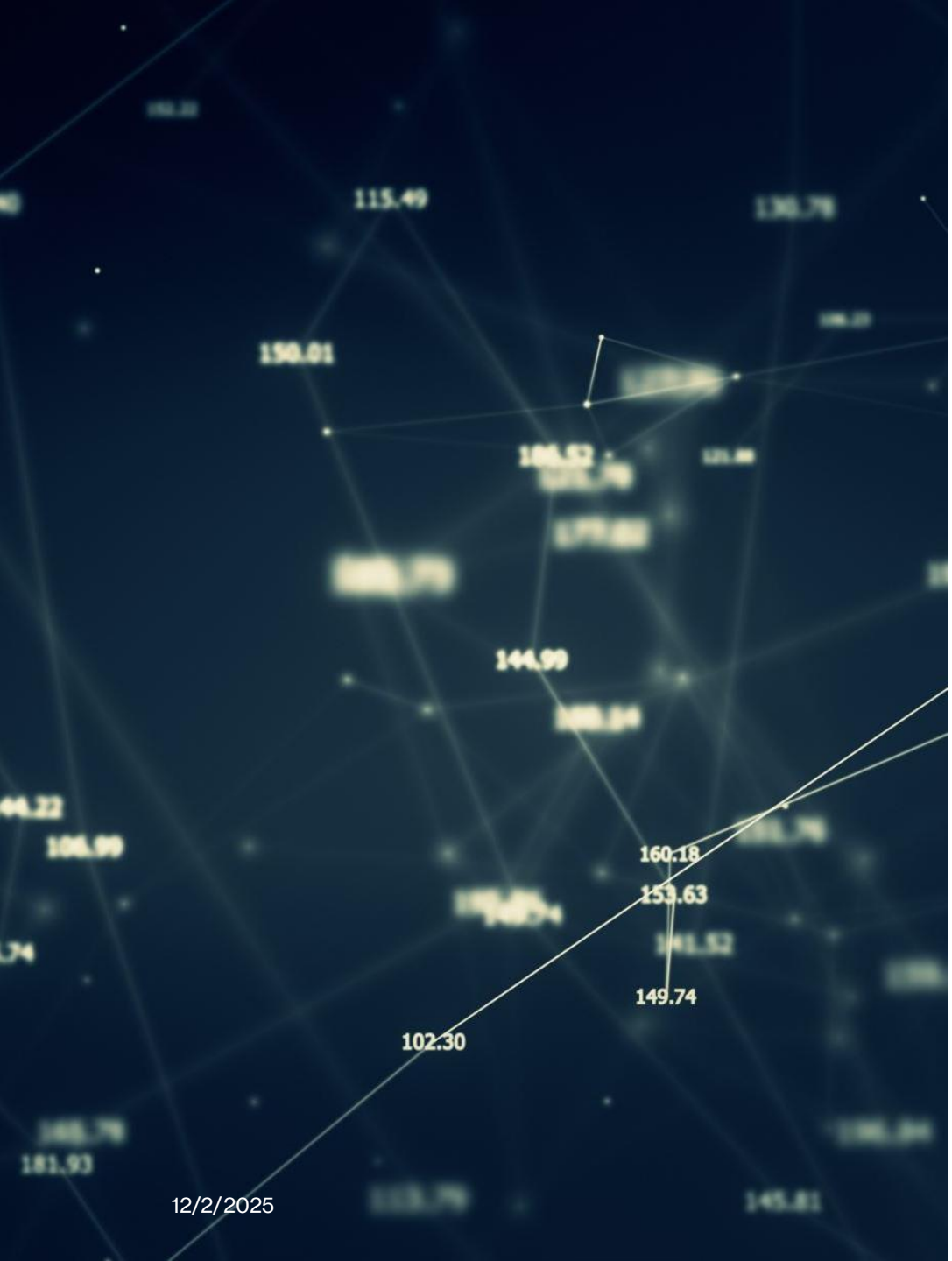
2. CSV files generated (sensor_data, events, potholes)

3. RealVNC enables remote access

4. Data uploaded to Supabase database

5. React website subscribes to changes

6. Dashboard updates within 1-2 seconds



DATABASE SCHEMA

- **Stored Data:**
- 1. **rides:** ride_id, start_time, duration, distance, final_score
- 2. **sensor_data:** timestamp, accel_xyz, gyro_xyz, speed, gps_coords
- 3. **events**: event_type, timestamp, severity, sensor_values
- 4. **pothole**s: timestamp, image_url, location, confidence
- Real-time subscriptions enable instant dashboard updates

WEB DASHBOARD (REACT + SUPABASE)

Real-Time Features:

3 Live Graphs: 3 acceleration + 3 gyroscope axes +speed

Speed Monitor: Current speed updating continuously

Event Timeline: Shows turns/stops as they happen

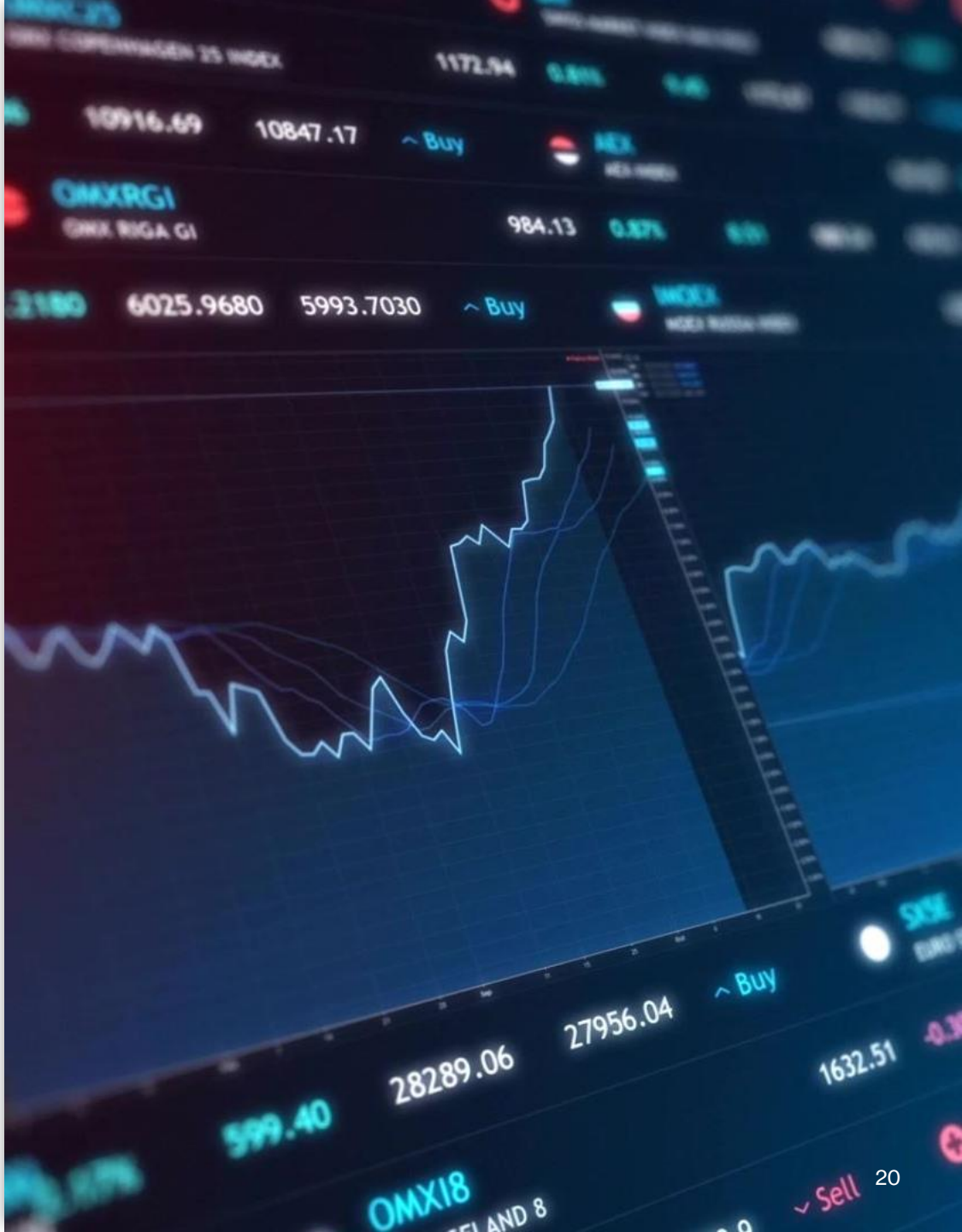
Pothole Alerts: Pop-up with timestamp when detected

Safety Score: Calculated automatically when ride ends

Technology: React.js frontend + Supabase database for real-time sync

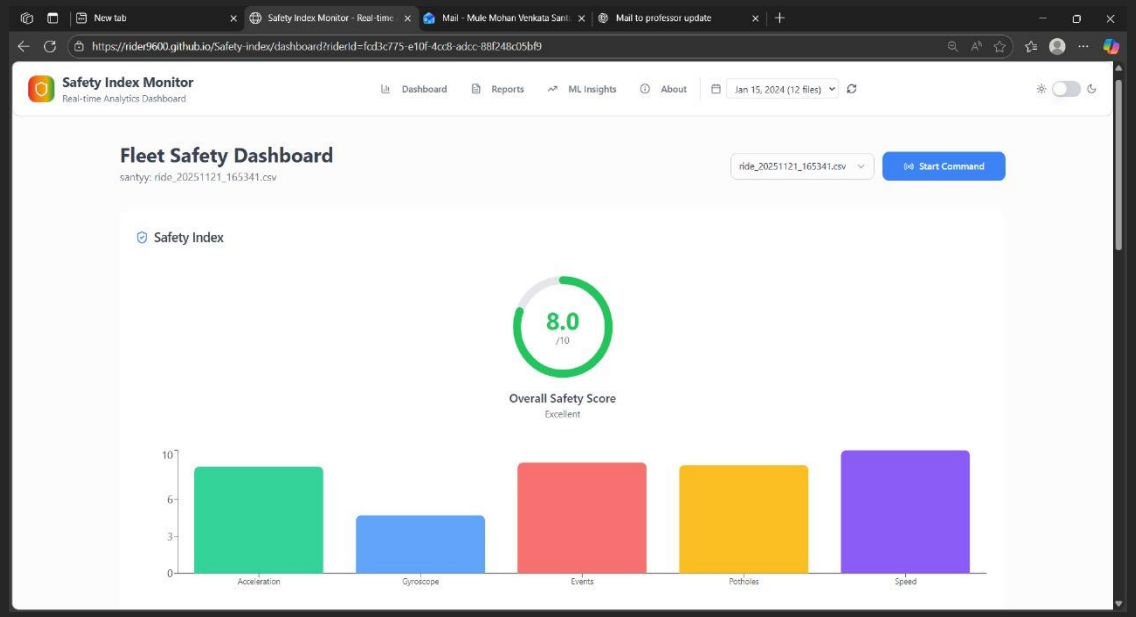
LIVE DASHBOARD FEATURES

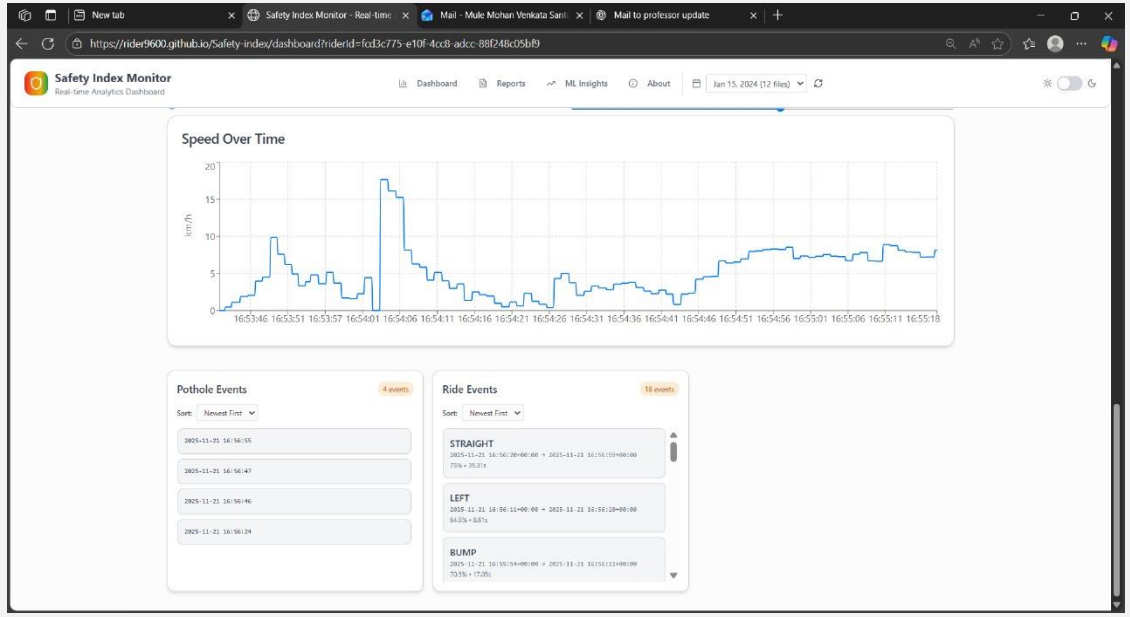
- **During Ride**
- • All 6 graphs updating continuously
- • Events appear on timeline instantly
- • If Pothole detected it shows in website that pothole is detected
- • Speed, duration
- **When Ride Stops:**
- • Final Safety Index displayed (e.g., 7.83/10)
- • Rating shown with stars
- • Complete ride statistics
- • Downloadable report

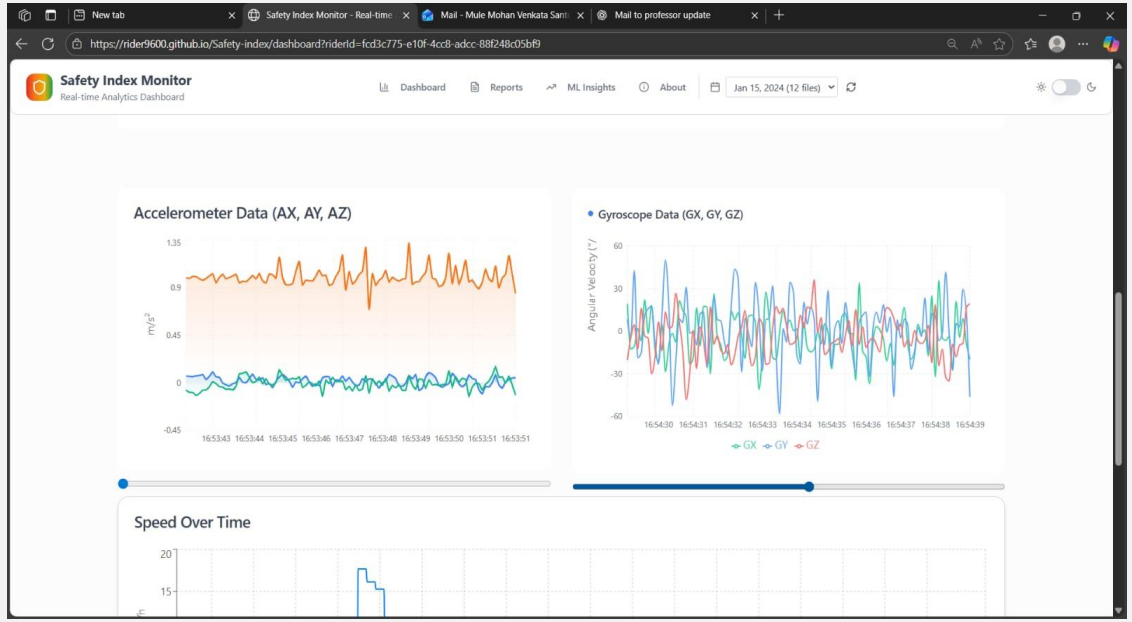


12/2/2025

WEBSITE







CAMPUS DEPLOYMENT

- **Testing:**
- • Deployed on scooter with 3D printed hardware box
- • Multiple 10-15 minute test rides
- • All data monitored live via website
- • Accessed remotely through RealVNC
- **Sample Results:**
- • Duration: 14.7 minutes • 92,000+ sensor samples
- • 154 harsh events detected
- • 2,908 aggressive turns
- • Final Score: 7.83/10 (GOOD)

CHALLENGES SOLVED

- 1. **GPS drift**: Applied Kalman filtering for smooth tracking
- 2. **False pothole alerts**: Added confidence threshold + shadow filtering
- 3. **Event classification noise**: Implemented 3-sample moving average
- 4. **Real-time sync delays**: Optimized Supabase queries to <2s latency
- 5. **Vibration interference**: Added hardware dampening in 3D box

MODEL PERFORMANCE METRICS

- **LSTM Event Detection:**
- • Training Accuracy: 81%
- • Real-time Inference: 15ms per prediction
- • Event Classification: 6 different riding patterns
- • Dataset: 100,000+ labeled sequences
- **ResNet-18 Pothole Detection:**
- • Overall Accuracy: 85%
- • Precision: 89% (low false positives)
- • Recall: 92% (catches most potholes)
- • Inference Speed: 200ms per image

APPLICATIONS

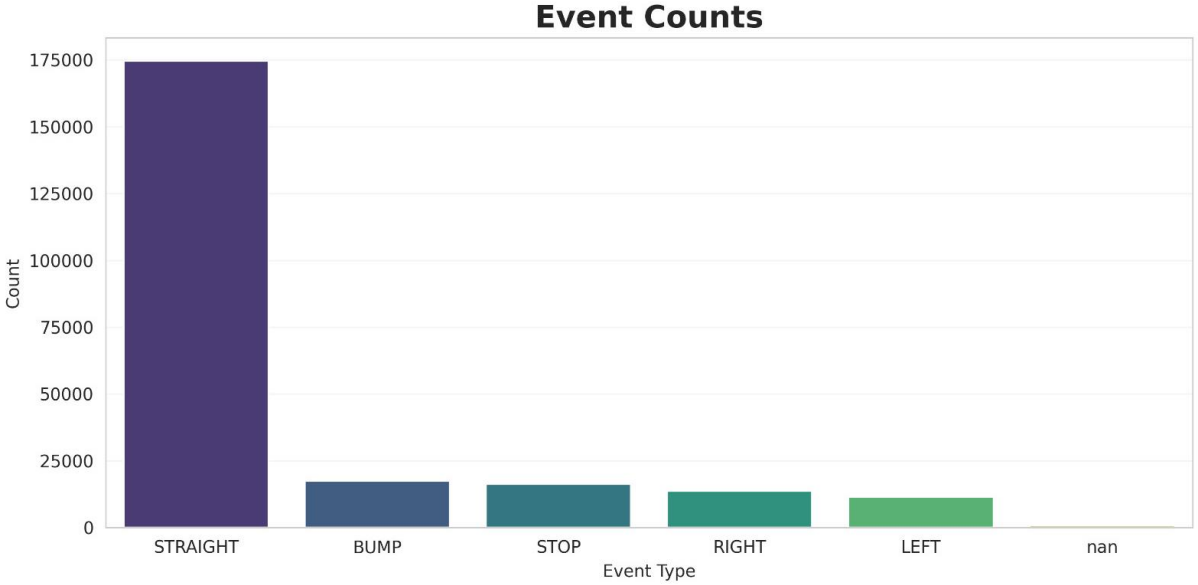
- **Use Cases:**
- Driver training with real-time feedback
- Fleet management for delivery services
- Insurance premium calculations
- City pothole mapping
- Accident prevention

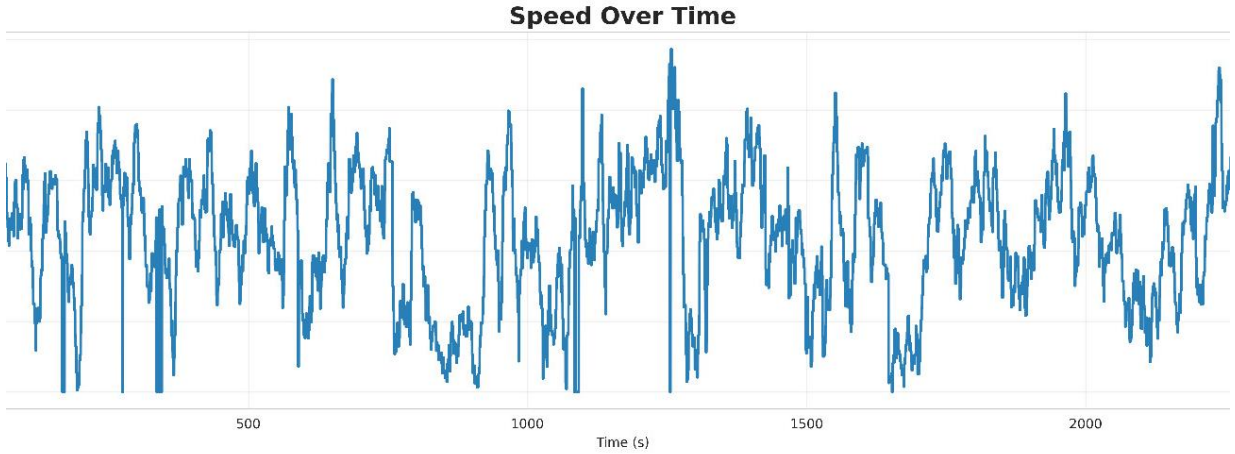
Testing outdoor

FUTURE ENHANCEMENTS

- Mobile app with push notifications
- Multi-vehicle fleet tracking
- Predictive safety alerts
- Social features and leaderboards
- Weather-adjusted scoring

Data


Event Counts
175000
150000
125000
100000
75000
50000
25000
0
Count
STRAIGHT
BUMP
STOP
RIGHT
LEFT
nan
Event Type


Speed Over Time
500
1000
1500
2000
Time (s)

CONCLUSION

- **We Built**: A complete real-time safety ecosystem combining IoT sensors, AI models, and live web visualization to make every ride safer.
- **Key Innovation**: Affordable, comprehensive, real-time monitoring with instant feedback.
- **Impact**: "Technology can save lives, one ride at a time."